TI Sigma Eight-C tests

Stage A record: what was run, what it used, and what comes next

Reading copy of analyses/eight_c_instrument_2026_10_20/results.json. Prepared 24 October 2026, America/Chicago. The checks themselves are dated 20–23 October 2026. Source of the instrument: product commit 58e78046101afdd58797e1f369f7b7ae4f204b5b (origin/product/truth-engine-alpha-1.1). Definitions status on that commit: NEW_CANONICAL_DRAFT. paid_collection_started: false.

Class labels used below. GENERATOR means the number describes the hash mock, not a human or LLM rater. SIMULATION means synthetic profiles drawn for a weight question. INSTRUMENT_TEXT means a comparison of two source files. BLOCKED means the registered input file could not be read. ABSENT means no rating table exists. A generator result is not a validation of the Eight Cs.

1. The precise definitions were not the scoring rule

The operational cards are eight_c_operational_definitions_v3.md. The rater prompt is ti_sigma_stage_a_v3_rater_prompt.txt. The mock reads the definition text: false. Its seed inputs are base_seed, item_id, scenario_text, attempt index, score name. Definition sentences contained unchanged in the prompt: 2. Reworded: 6. Operational questions absent: 8 of 8. High/mid/low anchor sets absent: 8 of 8. Do-not-confuse lines absent: 8 of 8. Goodness in the prompt is “Structural and evaluative adequacy expressed through the Eight Cs,” and the prompt also says to rate Goodness independently: true.

EXACT means the card’s definition sentence is contained in that C’s prompt block. REWORDED means the prompt uses a shorter or altered sentence. The six changes are: Coherence drops “stated aims.” Consistency changes “constraints, and” and “across” to “rules, or” and “within.” Continuity drops “developmental.” Completion drops “stages.” Closeness changes “adjacent” to “nearby.” Choice drops “evaluated.” Only Concreteness also carries its HEM boundary in the prompt.

C	Definition	Question	Anchors	Do not confuse	HEM boundary
coherence	REWORDED	ABSENT	ABSENT	ABSENT	ABSENT
consistency	REWORDED	ABSENT	ABSENT	ABSENT	ABSENT
continuity	REWORDED	ABSENT	ABSENT	ABSENT	ABSENT
concreteness	EXACT	ABSENT	ABSENT	ABSENT	PRESENT
completion	REWORDED	ABSENT	ABSENT	ABSENT	ABSENT
criticality	EXACT	ABSENT	ABSENT	ABSENT	ABSENT
closeness	REWORDED	ABSENT	ABSENT	ABSENT	ABSENT
choice	REWORDED	ABSENT	ABSENT	ABSENT	ABSENT

Canonical sentence and the prompt sentence

coherence (REWORDED). Card: degree to which parts, relations, and stated aims hang together as an intelligible whole. Prompt: Degree to which the parts and relations hang together as an intelligible whole.

consistency (REWORDED). Card: degree to which claims, constraints, and actions avoid contradiction across the relevant frame. Prompt: Degree to which claims, rules, or actions avoid contradiction within the relevant frame.

continuity (REWORDED). Card: degree to which the evaluated entity preserves intelligible linkage across time, stages, or developmental transitions. Prompt: Degree to which intelligible linkage is preserved across time, stages, or transitions.

concreteness (EXACT). Card: degree of tangibility or determinate intelligibility of the evaluated entity: how readily it can be understood, represented, or operationally grasped with minimal fuzziness, vagueness, or ambiguity. Prompt: **Concreteness is the degree of tangibility or determinate intelligibility of the evaluated entity: how readily it can be understood, represented, or operationally grasped with minimal fuzziness, vagueness, or ambiguity.** / Concreteness in GILE-G does not measure ontological instantiation. Ontological instantiation and concrete physical mechanisms belong to HEM.

completion (REWORDED). Card: degree to which the evaluated entity includes the parts, stages, or conditions needed for its stated function within scope. Prompt: Degree to which the entity includes the parts or conditions needed for its stated function within scope.

criticality (EXACT). Card: degree to which the evaluation appropriately tracks how much correctness, failure tolerance, and accuracy discipline matter for the case. Prompt: Degree to which the evaluation appropriately tracks how much correctness, failure tolerance, and accuracy discipline matter for the case.

closeness (REWORDED). Card: degree to which the evaluated entity remains near the relevant target, need, or purpose rather than drifting into adjacent but less relevant success. Prompt: Degree to which the entity remains near the relevant target, need, or purpose rather than drifting into nearby but less relevant success.

choice (REWORDED). Card: degree to which the relevant bearer has meaningful, attributable, and normatively interpretable choice within the evaluated situation, including the alignment of that choice with the relevant target. Prompt: Degree to which the relevant bearer has meaningful, attributable, and normatively interpretable choice within the situation, including the alignment of that choice with the relevant target.

2. Generator reproducibility

The stored file results/reports/ti_sigma_stage_a_v3_mock_reproducibility.json matches vary_by_attempt with base seed TI_SIGMA_STAGE_A_V3_MOCK: true. The registered config seed TI_SIGMA_STAGE_A_V3_REGISTERED matches that file: false. Item groups 21. Attempts per item 3. Exact-match rate 0.0. Goodness mean absolute difference 4.031746031746032. Registered-seed goodness mean absolute difference 4.317460317460318. The fixed strategy on the CLI seed returns exact-match 1.0 and goodness mean absolute difference 0.0.

Expected absolute difference of two independent uniform draws on {0..10} is 440/121 (440/121). On {0..3} it is 20/16 (20/16). Mean of the eight C gaps: 3.511904761904762, uniform two-sided tail 0.72595 from 20000 draws, seed 20261021. Goodness gap tail 0.2897. Mean contradiction gap 1.2936507936507937, tail 0.71375. choice_bearer and choice_scope are not hash inputs. Substituting them left the scores unchanged: true. The items CSV and the metadata CSV were not downloaded. The repository LFS budget was exceeded.

Pearson r of the separate goodness bucket with the mean of the eight Cs, 63 mock rows: 0.012507296714551402. Permutation p 0.9198160367926415 from 5000 shuffles, seed 20261022.

Score	CLI-default mean abs. diff.	Registered-seed mean abs. diff.
goodness	4.031746031746032	4.317460317460318
coherence	3.4603174603174605	3.8730158730158726
consistency	3.746031746031746	3.746031746031746
continuity	3.142857142857143	3.555555555555556
concreteness	3.587301587301587	3.619047619047619
completion	3.5238095238095237	3.4603174603174605
criticality	3.4603174603174605	3.7777777777777777
closeness	3.3968253968253967	3.8095238095238093
choice	3.7777777777777777	3.8412698412698414

Contradiction	CLI-default mean abs. diff.
category_error	1.2698412698412698
competing_domains	1.2063492063492063
genuine_tradeoff	1.3015873015873016
incomplete_information	1.4285714285714286
measurement_uncertainty	1.2380952380952381
self_reference	1.3015873015873016
temporal_inconsistency	1.26984126984127
true_incoherence	1.3333333333333333

3. Corpus balance

Status: INCONSISTENT. JSONL items: 21. Cluster roles A/B/C for clusters 1–7 complete: true. Stored domain counts sum to 23. Stored scale counts sum to 21. Domains match the JSONL: false. Scales match the JSONL: false. The balance report is prose on the product commit. No script in that commit writes it. The item CSV is an unread LFS pointer, so this comparison uses the JSONL.

Domain	JSONL	Stored prose
AI	3	3
biology	2	2
engineering	3	4
ethics/economics	1	1
individual cognition	2	2
institutions/governance	3	3
interpersonal relations	2	2
mathematics/formal reasoning	1	1
science	4	5

Scale	JSONL	Stored prose
biological	2	2
cognitive	3	3
formal	1	1
institutional	6	5
interpersonal	2	2
physical	4	5
societal	3	3

4. Schema, moments, and agreement

Schema records 63. Failures 0. Status REPRODUCIBLE. Class GENERATOR. Fixed-strategy goodness ICC(1,1) 1.0. ICC(2,1) 1.0. The vary-by-attempt moments below are 63 hash scores. The uniform mean on {0..10} is 5. Goodness ICC(1,1) versus 2000 independent uniform studies, seed 20261023: Monte Carlo mean -0.0028445151813530963, two-sided tail 0.5065.

Score	Mean	Sample SD	ICC(1,1)	ICC(2,1)
goodness	4.523809523809524	3.3207905338315538	-0.0883895491507482	-0.10209749323978659
coherence	5.349206349206349	3.13234288362628	0.03528144357662260 6	0.02195281782437744
consistency	5.285714285714286	3.013028545139406	-0.1727128875344313	-0.19096509240246415
continuity	4.523809523809524	2.8672631823189914	0.11515773275198764	0.1120548209632274
concreteness	3.9682539682539684	3.3067294726704732	0.02977859421831179 4	0.03211805555555543 5
completion	5.238095238095238	3.078092652582145	0.06202976090954693	0.08676542405990564
criticality	5.285714285714286	3.347741518081915	0.14865626761886852	0.16924558145932184
closeness	4.2063492063492065	3.13234288362628	0.1284034154986305	0.12032520325203239
choice	4.809523809523809	3.2915196159963074	-0.07961222091656874	-0.0963603818615752

5. Registered attempt accounting

Original function ti_sigma_pipeline.stage_a_v3._mock_response_for_attempt. Seed strategy vary_by_attempt. Base seed TI_SIGMA_STAGE_A_V3_REGISTERED. Planned logical ratings 63. First-attempt valid 58 (rate 0.9206349206349206). Eventual valid 63. Eventual invalid 0. Total attempts 68. Cap 189. Inside the cap: true. Retry keys: registered_stage_a:V3-03:2, registered_stage_a:V3-07:1, registered_stage_a:V3-11:3, registered_stage_a:V3-16:2, registered_stage_a:V3-20:1. The five keys fail the first attempt because the mock inserts an unknown property, then the second attempt is valid.

6. Do the eight scores share a common construct?

Unit: 21 item means of 3 hash attempts. Null: the eight columns are mutually independent. Statistic: the largest Pearson eigenvalue. 5000 column shuffles, seed 20261024. Reject when p < 0.05. First eigenvalue 2.2853934527091626. Permutation p 0.14457108578284344. Independence rejected: false. Cronbach alpha 0.146050066723026. Effective rank 6.58819161744164 on a scale from 1 to 8. Horn parallel analysis, 2000 shuffles, seed 20260902: retained components 0. PC1 share of variance 0.2856741815886454. Residual share after PC1 0.7143258184113546. Mean PC1 communality 0.2856741815886453. Status: NO_COMMON_CONSTRUCT. One common construct accounts for the eight scores: false. N=21 is diagnostic. The preregistration forbids a strong completeness claim from 21 items.

Component	Observed eigenvalue	Horn 95th percentile
1	2.2853934527091626	2.44626277509581
2	1.566472412942264	1.8368005879573202
3	1.2776758930334626	1.4675463122095032
4	0.9898575949459065	1.1874527116707114
5	0.6777160478997316	0.9544237123755368
6	0.504163993821385	0.7465686524037807
7	0.37582582672455034	0.5534406410188998
8	0.3228947779235356	0.367465074044578

C	PC1 communality
coherence	0.0001316209687222095
consistency	0.38868054085042947
continuity	0.03625831819671152
concreteness	0.1437092248588136
completion	0.45507247437773574
criticality	0.4748920924039569
closeness	0.40495455314831935
choice	0.381694627904474

7. Does that construct, or Goodness, get fully accounted for by the eight Cs?

There is no retained common component, so there is no common factor for the eight scores to exhaust. The descriptive PC1 share above is the share of a component that did not clear the null. Goodness is a separate hash key. Full linear accounting was defined in advance as leave-one-out R-squared equal to 1. In-sample R-squared 0.3019137838619306. Permutation p 0.7320535892821436. Leave-one-out R-squared -1.2630063958020945. Fully accounted for: false. Incremental Choice delta R-squared 0.007453560096458811. Permutation p 0.7280543891221756. Draws 5000, seed 20261024. Gaussian CMI from partial correlations. N=21 with seven conditioners is unstable and is not a confirmation sample.

Dropped C	In-sample R^2 without it	Delta R^2
coherence	0.30177648044772554	0.00013730341420503667
consistency	0.21243729637018505	0.08947648749174553
continuity	0.20804685282090596	0.09386693104102461
concreteness	0.30175027592432435	0.00016350793760622206
completion	0.30087718297883737	0.0010366008830932083
criticality	0.1414905151823206	0.16042326867961
closeness	0.297597513037685	0.004316270824245572
choice	0.29446022376547176	0.007453560096458811

C	Partial r with Goodness	Gaussian CMI (bits)
coherence	-0.014023080636859827	0.00014186462383376138
consistency	-0.3370636294578224	0.08699478307945803
continuity	-0.3442758545562487	0.09100492798908631
concreteness	0.015302557989283138	0.00016893648943290 84
completion	0.03850605697628725	0.001070347523037781
criticality	-0.43227600751979683	0.1492144238977887
closeness	-0.0783901188750032	0.00444636376977501

C	Partial r with Goodness	Gaussian CMI (bits)
choice	0.10278296156996723	0.007661097274351489

8. Named pairs, contradictions, and a mock kappa

Exchangeability of the three named pairs in the mock item means. Six p-values, no multiplicity adjustment. A single unadjusted p-value is not the registered directional contrast. Engineered directional contrast: BLOCKED.

Pair	Mean difference	p	Pearson r	p
choice_versus_concreteness	0.8412698412698413	0.14557088582283542	0.08257557576855584	0.7438512297540492
closeness_versus_continuity	-0.3174603174603175	0.5720855828834233	0.15990309461929067	0.4801039792041591 5
concreteness_versus_completion	-1.2698412698412698	0.04999000199960008	-0.018834432613821724	0.9488102379524095

Contradiction peak-minus-rest mean 1.63718820861678 on 63 mock rows. One-sided permutation p 0.22975404919016196 against eight independent uniform{0,1,2,3} contradiction scores. Designed type selectivity: BLOCKED.

Fleiss kappa treating the three hash attempts as raters and integer scores 0–10 as categories. A human or LLM kappa: ABSENT.

Score	Fleiss kappa
goodness	-0.0298759864712514
coherence	0.04276871131119865
consistency	0.0011325028312570737
continuity	-0.04586381541924592
concreteness	-0.017888055395268346
completion	-0.051187782805429856
criticality	-0.0008361204013377816
closeness	0.005076142131979691
choice	-0.004756575265808607

9. Weight simulation

This check is not a Stage A primary analysis. It follows the owner’s statement that the weights are deliberately unclear. Seed 20261020. Matrices 200. Items 24. Scores uniform integers {0..10}. Proportional profiles, every equal-weight versus choice-only Kendall tau-b equals 1: true. Independent mean Kendall tau-b 0.24050465826369907. Minimum -0.11315527740769757. Fraction below 1: 1.0. Duplicate-column maximum absolute composite difference 0. Constructed transfer onto concreteness, Kendall tau-b -1.0. Domain weights assigned: false. Closest applicability: not scored; no distance was defined.

10. What comes next

The registered directional contrasts and the designed contradiction types are columns of the metadata CSV. Git LFS returned the budget error and the object was not downloaded. No provisional profile was inferred from item titles. File: experiments/eight_c_goodness_pilot/data/metadata/ti_sigma_stage_a_v3_metadata.csv. LFS oid sha256:f5f3a276063af7f895ac4f61c49c349bb02cfbcc50327a9bd69a7819221f82e1. Size 4549 bytes. The unread columns are intended_coherence, intended_consistency, intended_continuity, intended_concreteness, intended_completion, intended_criticality, intended_closeness, intended_choice, intended_goodness, intended_contradiction_types, intended_contrast, predicted_profile_notes. The prompt also requires choice_bearer and choice_scope, and both are in that same file. A rating pass was not started, and no profile was inferred from item titles.

The next tests, in the order the instrument requires them:

1. Read the metadata CSV once LFS access is restored. Confirm its sha256 is f5f3a276063af7f895ac4f61c49c349bb02cfbcc50327a9bd69a7819221f82e1 before using any intended profile.

2. Run the directional A/B/C contrasts against intended_contrast and the intended_* score columns. Do not infer those directions from item titles.

3. Run designed contradiction-type selectivity against intended_contradiction_types.

4. Decide the rater text before any new kappa. The current prompt is a shortened gloss. A study that is compared with the v3 cards needs the definition sentence, the operational question, the three anchors, and the do-not-confuse line written into the instructions.

5. Collect the 63 logical ratings under the registered stopping rule. paid_collection_started is false, and that collection was not authorized by the preregistration's local-verification boundary.

6. Estimate a human or LLM Fleiss kappa, the three named discriminations, and a common-factor test on those ratings. The hash results in this record do not transfer.

7. Keep Goodness as its own rating. Ask again whether the eight Cs account for it. Full linear accounting, as defined here, is leave-one-out R-squared equal to 1.

8. Leave N=21 diagnostic. The preregistration forbids a strong completeness claim from 21 items.

Files. results.json is the numeric source. EIGHT_C_INSTRUMENT_REPORT.md is the short narrative. run_eight_c_instrument.py and stage_a_remaining.py recompute both. tests/test_eight_c_instrument.py locks the JSON. This PDF is produced by build_test_record_pdf.py from results.json.